

ADVANCED NEUROLOGY CLINICAL TELEMONITORING CENTER

CONFIDENTIAL MEDICAL RECORD
Ref ID: #PARK-2026-884
Date: September 28, 2026

Department of Neurodegenerative Disorders & Motor Rehabilitation

Protocol: MDS-UPDRS Phonation Screening

Patient Name: Arthur Vance	Patient ID: PT-94021	Age: 68
Biological Sex: 1 (Male)	Assessment Type: Voice & Motor Telemetry	Clinician: Dr. Eleanor Vance, MD

1. Motor Performance & Clinical UPDRS Baseline

The patient completed the standardized Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS) Part III motor assessment. Recorded baseline evaluation indicates mild-to-moderate motor symptoms including unilateral resting tremor, mild resting rigidity, and bradykinesia during finger-tapping trials.

Clinical Measurement	Measured Value	Standard Clinical Scale	Evaluation Status
MDS-UPDRS Part III Motor Score (motor_UPDRS)	UPDRS: 22.4	0 – 108 scale	Mild-to-Moderate (Stage 2)
Patient Chronological Age (age)	age: 68	Years	Target Diagnostic Cohort
Biological Sex (sex)	sex: 1	1 = Male, 0 = Female	Validated

2. Acoustic Phonation Biomarkers (12-Feature ANOVA Model Profile)

Sustained phonation of the neutral vowel /a/ recorded at 44.1 kHz, 16-bit uncompressed audio in a sound-attenuated clinical telemetry chamber. Pitch tracking and perturbation metrics were extracted using algorithmically calibrated MDVP (Multidimensional Voice Program) and Praat-compatible acoustic formulations.

Acoustic Biomarker	Standard Metric ID	Measured Value	Reference Cohort Range
MDVP Relative Jitter	Jitter(%)	Jitter(%): 0.0054	< 0.0040 (Normal)
MDVP Absolute Jitter	Jitter(Abs)	Jitter(Abs): 0.000042	< 0.000030 s
Relative Average Perturbation	Jitter:RAP	Jitter:RAP: 0.0028	< 0.0025
5-Point Period Perturbation Quotient	Jitter:PPQ5	Jitter:PPQ5: 0.0031	< 0.0030
Average Degree of Period Difference	Jitter:DDP	Jitter:DDP: 0.0084	< 0.0075
Harmonics-to-Noise Ratio (dB)	HNR	HNR: 21.65	> 20.00 dB (Desirable)
Detrended Fluctuation Analysis	DFA	DFA: 0.658	0.50 – 0.85 scale
Recurrence Period Density Entropy	RPDE	RPDE: 0.542	0.00 – 1.00 dynamical complexity
Pitch Period Entropy	PPE	PPE: 0.231	0.00 – 0.60 logarithmic entropy

CLINICAL INTERPRETATION & DECISION SUPPORT SUMMARY:

- **Vocal Dysphonia:** Elevated micro-perturbation ratios (Jitter: % = 0.0054, Jitter:RAP = 0.0028) with conserved harmonic ratio (HNR = 21.65 dB) consistent with early laryngeal rigidity.
- **Entropy Dynamics:** Nonlinear dynamical indicators (RPDE = 0.542, PPE = 0.231, DFA = 0.658) indicate emergent vocal-tract tremor and periodicity destabilization.
- **Machine Learning Compatibility:** Contains all 12 ANOVA-selected feature dimensions for direct ingestion by Quantum Neural Network (QNN Phase 24), Hybrid VQC, and Federated Random Forest models.

Examining Specialist:

Eleanor Vance, MD, PhD
Lead Neurologist, Telemonitoring Unit

Facility Certification:

Certified ISO/IEC 13485 Clinical Health
Informatics
MDS-UPDRS Certified Telehealth Rater

Report Status:

VERIFIED & SIGNED
Digital Stamp #89CC-CD01